🧠 Thalamus + Internal Capsule: Cram Sheet

🚉 Big rules

- **Grand Central Station:** every sense relays here on the way to cortex **except smell**. Taste does (VPM).
- Thalamus = **3rd-order neuron** of sensory pathways.
- **Front-to-back rule:** front of thalamus → front of cortex → front of capsule. Back → back → back. **Ant**→cingulate · **VA**→premotor · **VL**→motor · **VPL/VPM**→somatosensory · **Pulv/LG**→back · **MD**→prefrontal
- **Anterior + medial = OLD** = emotion, visceral, limbic. **Lateral = NEW** = sensory + motor (huge).
- Lesion = **contralateral** body signs (tracts already crossed).
- **More than a relay:** "silos". Each nucleus keeps its info sealed. Details of the silo diagram NOT needed.

🗺️ Layout

Massa intermedia (IA)	joins the 2 thalami across 3rd ventricle; on **medial** surface
Internal medullary lamina	Y-shaped white sheet; anterior sits in the fork
Intralaminar n.	inside the lamina (intra = inside)
Reticular n.	thin net/shell outside external lamina, like orange peel
Tiers	dorsal tier on top, ventral tier below
Geniculate = knee	2 bumps at the **back**: MG medial, LG lateral
DM = MD	same nucleus, two names

Names are directions: **V**entral **A**nterior, **V**entral **L**ateral, **V**entral **P**osterior **L**ateral / **M**edial.

🔌 Nuclei tables (he may blank a cell)

In	Nucleus	Out	Job
Sensory			
Dorsal column / spinothalamic	**VPL**	Postcentral	Body sens.
Trigeminal	**VPM**	Postcentral	Face sens.
Inferior colliculus	**MG**	Temporal	Hearing
Retina (optic tract)	**LG**	Occipital	Vision
Motor (extrapyramidal)			
Cerebellum	**VL > VA**	Frontal	Motor plan
Globus pallidus	**VA > VL**	Premotor	Motor plan
Association			
Hypothalamus (mammillothalamic)	**Anterior**	Cingulate	Limbic
Amygdala	**MD**	Prefrontal	Limbic
Anterolateral (= spinothalamic)	**Posterior**	Cingulate + assoc.	Pain
Superior colliculus	**Pulvinar**	Temp/par/occ	Vision
Nonspecific (FYI)			
Reticular formation	**Midline**	Entire cortex	Arousal
Motor cx / BG	Intralam.	Striatum + cx	Motor integ. (he skips)
Cortex + brainstem	**Reticular**	**Thalamus**	Modulation

🧩 **M**G = **M**usic · **L**G = **L**ight · VP**L** = **L**imbs · VP**M** = **M**outh

✏️ Blank thalamus diagram (slide 19)

- **Step 1:** left or right? Find midline side + bumps.
- Bumps (geniculates) = **back**. MG medial, LG lateral.
- Y lamina: anterior in the fork; medial on one side, lateral on the other.
- Ventral tier front→back: **VA, VL, VPL**, VPM tucked behind.
- Dorsal tier: **LD, LP**, then **pulvinar** at back.
- Slide 18 blanks: **visual cortex** (LG) + **auditory cortex** (MG).

🔴 If I see X → I answer Y

Pure **sensory** loss, face + arm + trunk + leg, one side, no weakness	→ contralateral thalamus, **VPL + VPM**
Pure **motor**, face + arm + leg equal, forehead spared	→ contralateral **internal capsule**
Months after thalamic stroke, light touch feels like a knife; exam unchanged	→ **Dejerine-Roussy** (failure of silo isolation)
Pain triggered by music	→ silos crossed: **MG** into pain
Weird organized visions, hallucinations, sleepy	→ **pulvinar** ("vision, but not vision")
Coma, lesion in central core of thalamus/brainstem	→ **RAS** (midline nuclei)
Staring spells, EEG **exactly 3/sec**	→ absence seizure, **reticular n.** takes over rhythm
Movement disorder (posturing, jerks) after thalamic stroke	→ **VL/VA** (BG + cerebellum route)
Poor judgment / planning	→ **MD** → prefrontal
Face numbness only	→ **VPM**
HTN + DM + tiny deep stroke	→ **lacunar infarct**

⚠️ Traps

- "All senses use thalamus" ✗ **smell skips**.
- Norma = VPL only ✗ face → **VPL + VPM**.
- Norma = new stroke ✗ exam **unchanged**.
- Pulvinar ≠ LG. LG lesion = **blind field**; pulvinar = **hallucinations**.
- Reticular **nucleus** (thalamic shell, absence) ≠ reticular **formation** (brainstem, arousal).
- Reticular n. output → **thalamus**, not cortex.
- Cerebellum → **VL**; pallidus → **VA**. Don't swap.
- Pure motor → capsule; pure sensory → thalamus.
- 3 "genus": capsule genu (face), geniculate bodies, corpus callosum genu.
- Delay of months does **not** rule out the old stroke.
- Absence = **exactly 3**/sec (cortex normally ~12).

🚨 Cases

- **Norma, 75F:** right face/arm/trunk/leg numb → **left VPL + VPM**. Months later touch hurts, screams: Dejerine-Roussy. Emotional because pain → posterior n. → **cingulate** (limbic).
- **Dejerine-Roussy 5:** pure sensory loss, no weakness · contralateral loss · contralateral pain · pain to light touch or none · worse with heat, cold, stress, **music**.
- **32F:** L-hand posturing, jerky rhythmic, action-worse, 3 mo after **R ventrolateral** thalamic stroke.
- **20M:** hallucinations (naked women, birds), slept 30 h → **L pulvinar** infarct.
- **Case 10.7, 84F:** HTN, DM, 2 TIAs then dysarthria + R hemiplegia, forehead spared → **L internal capsule** (posterior limb).
- **Haley:** absence seizure; danger crossing the street.

🛣️ Internal capsule (front → back)

Anterior limb	anterior thalamic radiation, frontopontine
Genu (knee, V to midline)	**corticobulbar = FACE**
Posterior limb	corticospinal **Arm → Trunk → Leg**, then sensory (superior thalamic radiation)
Behind lentiform	auditory radiation, **optic radiation**

Capsule = **axons only**. Corona radiata = crown of fibers fanning to cortex.
Why not cortex? MCA = face + arm, ACA = leg. One cortical stroke cannot hit all three.